Model Card: Adult Income Classifier

Model details

A Logistic Regression classifier trained on the Adult Income dataset.

Intended use

Educational demonstration of machine learning evaluation and fairness. This model is not intended for real employment, lending, or income decisions.

Training and evaluation data

The model uses the Adult Income benchmark dataset. The data was split into training and test sets using an 80:20 ratio.

Quantitative results

Accuracy: 0.8523

Prediction disparate impact: 0.3238

Prediction statistical parity difference: -0.1660

Ethical considerations and limitations

The dataset reflects historical income patterns and may contain structural inequalities. Fairness metrics capture only selected aspects of fairness. Results should be examined across groups, and SHAP explanations do not establish that a model is fair. The model must not be used to make real-world decisions about individuals.